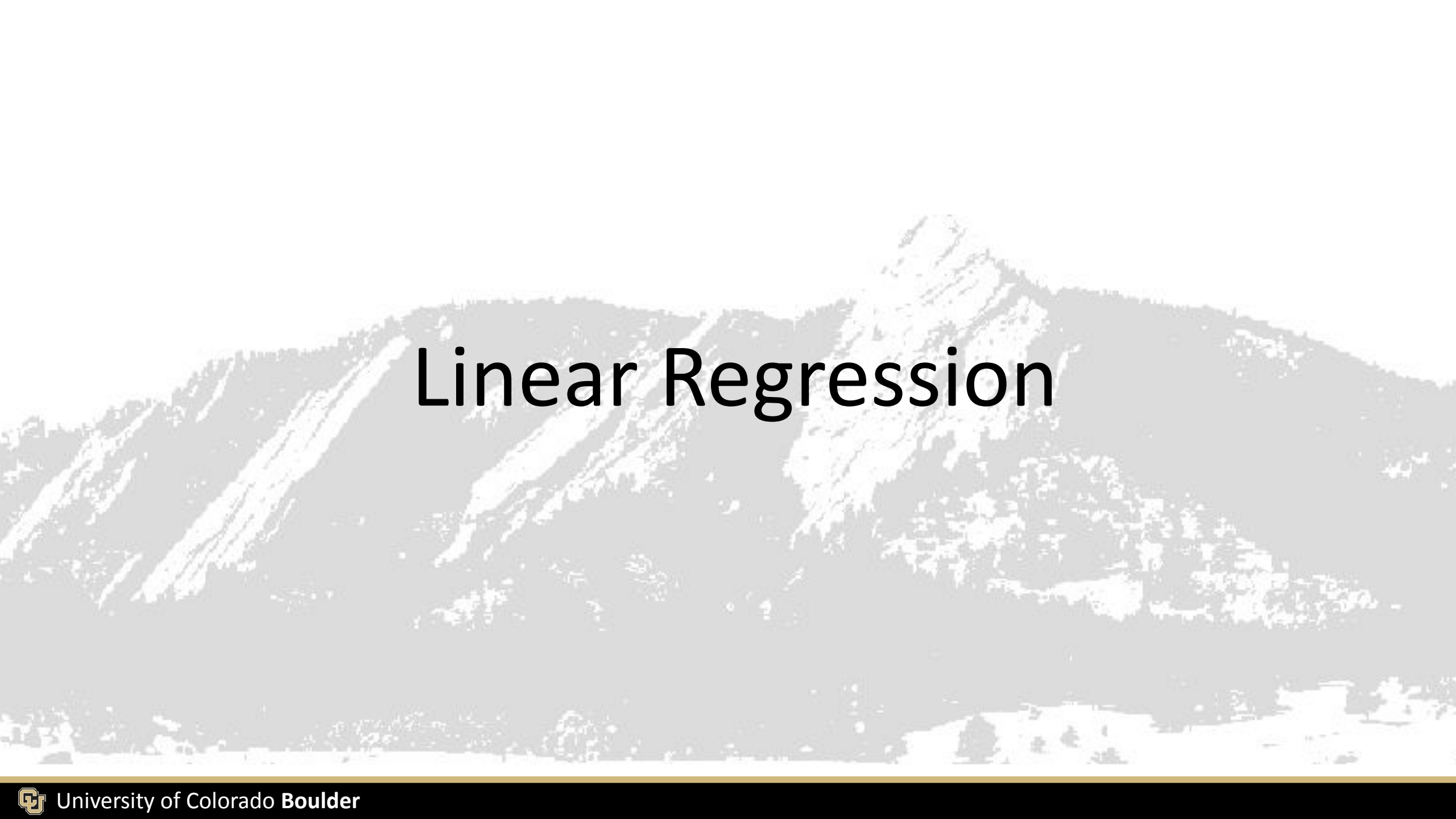
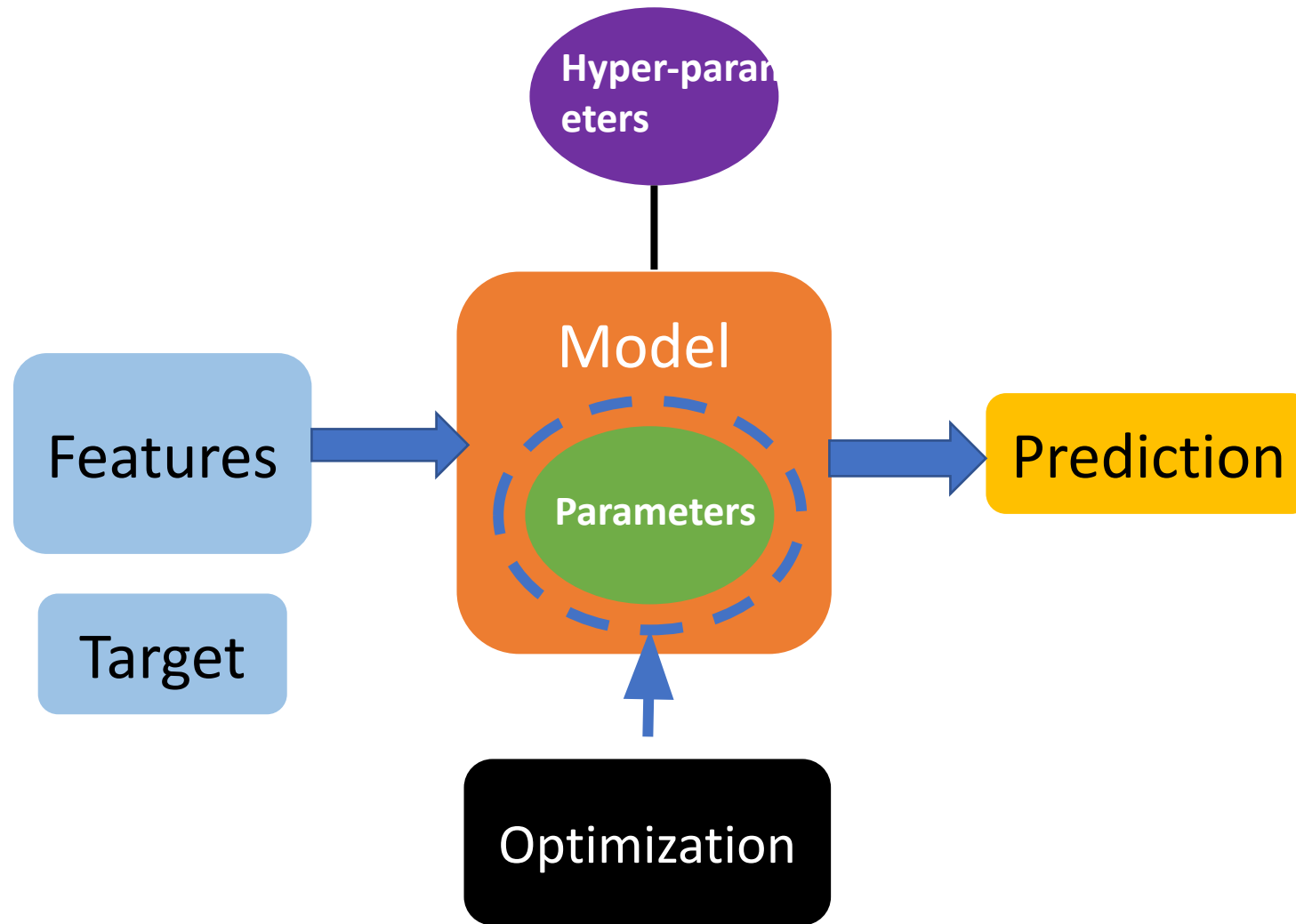




Linear Regression

How Supervised Learning Works



What is Linear Regression

- Supervised learning model
- Predictive task- real valued numbers
- Parametric model
- No hyperparameters
- Features have linear relationship to the target variable

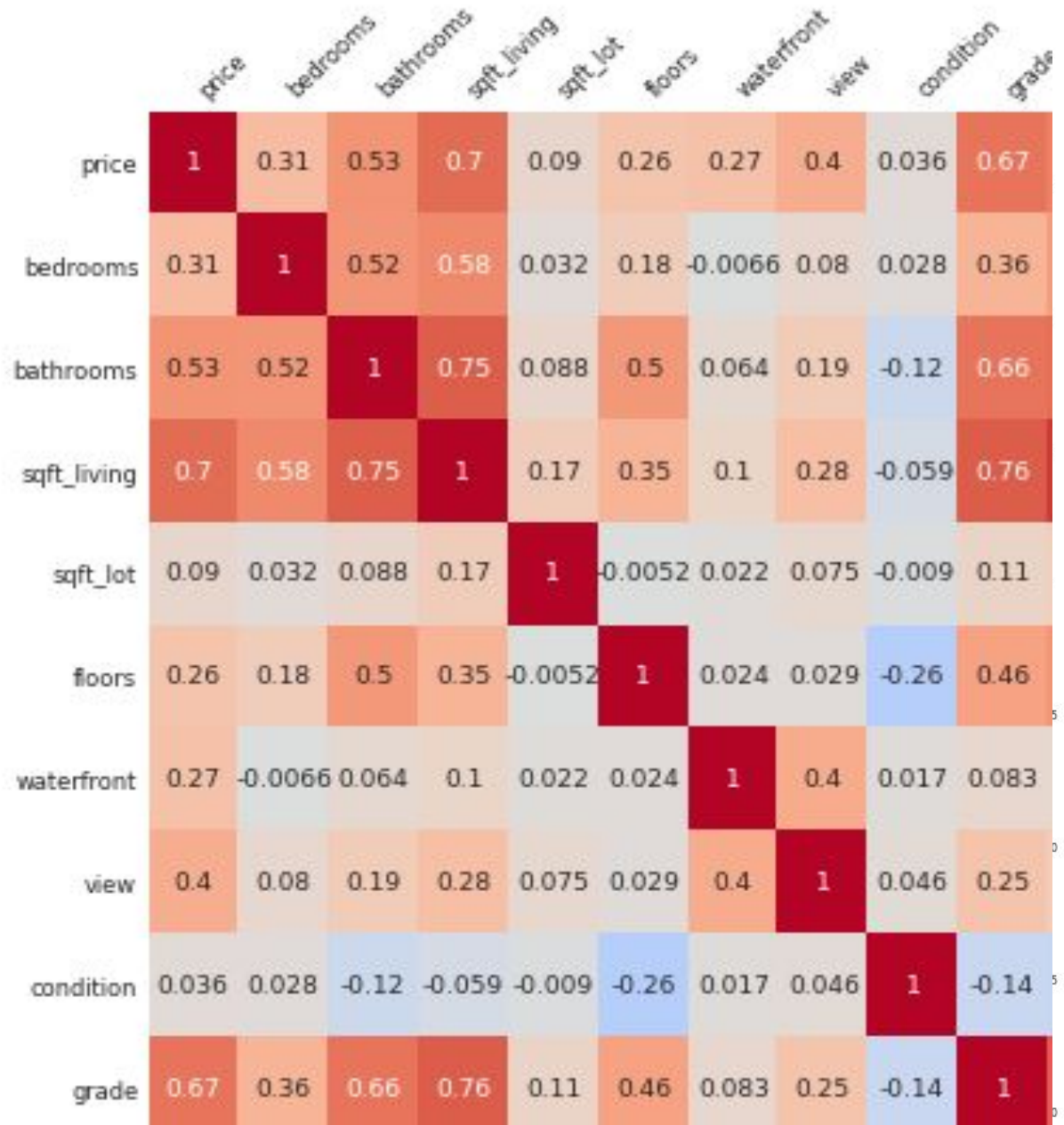
Example

An example using House sales data from Kaggle

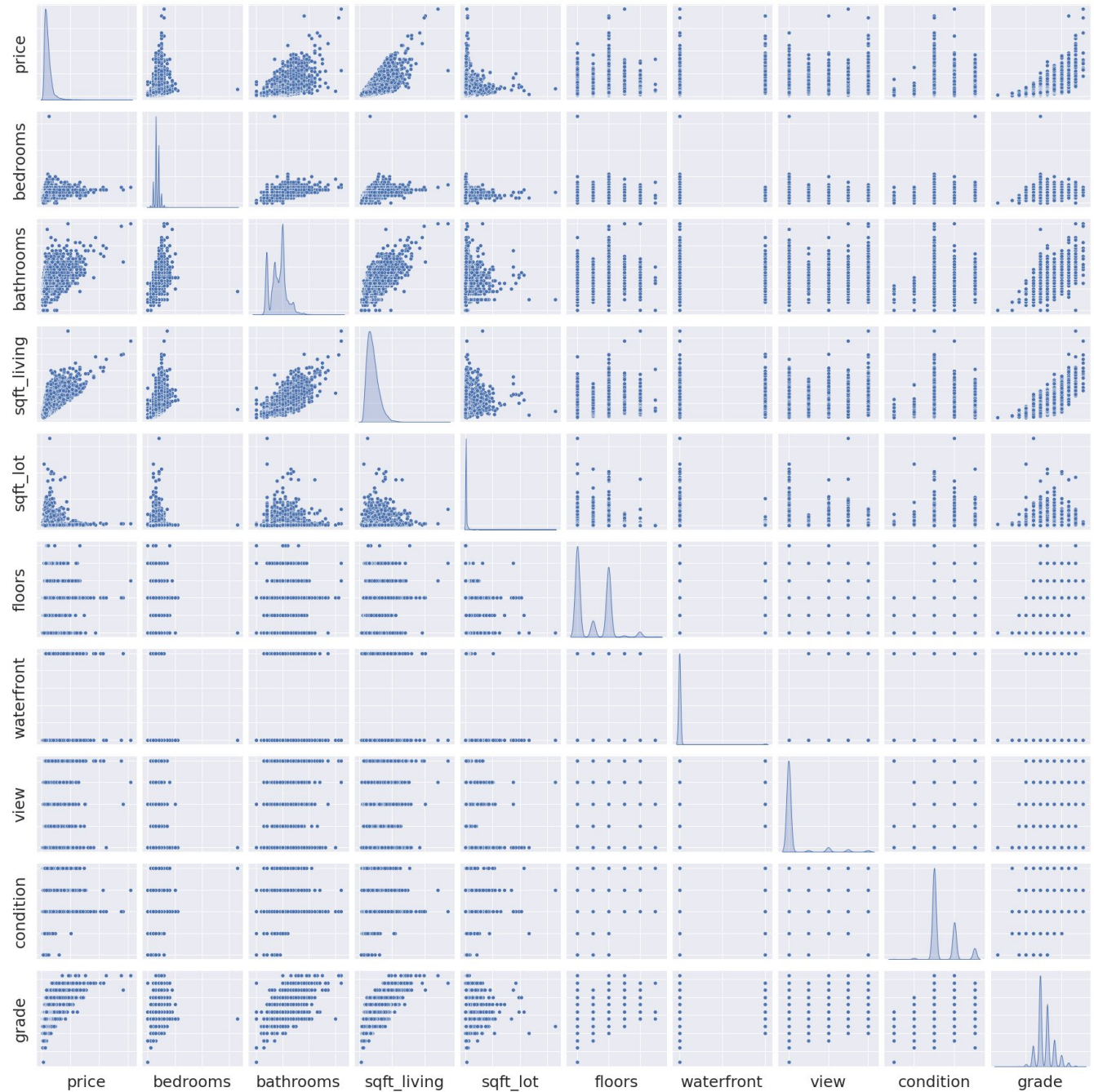
<https://www.kaggle.com/harlfoxem/housesalesprediction/download>

price	bedrooms	bathrooms	sqft_living	sqft_lot	floors	waterfront	view	...	grade	sqft_above	sqft_basement	yr_built	yr_renovated	zipcode
221900	3	1.00	1180	5650	1.0	0	0	...	7	1180	0	1955	0	98178
538000	3	2.25	2570	7242	2.0	0	0	...	7	2170	400	1951	1991	98125
180000	2	1.00	770	10000	1.0	0	0	...	6	770	0	1933	0	98028
604000	4	3.00	1960	5000	1.0	0	0	...	7	1050	910	1965	0	98136
510000	3	2.00	1680	8080	1.0	0	0	...	8	1680	0	1987	0	98074

Correlation Matrix



Pair Plot



Univariate Linear Regression

$$Y = \underbrace{\beta_0}_{\text{Intercept}} + \underbrace{\beta_1}_{\text{Slope}} X + \underbrace{\epsilon}_{\text{Residual}}$$

Coefficients, or Parameters

Using statsmodel's OLS (ordinary least squares) package



OLS Regression Results

Dep. Variable:	price	R-squared:	0.493			
Model:	OLS	Adj. R-squared:	0.493			
Method:	Least Squares	F-statistic:	2.100e+04			
Date:	Thu, 25 Feb 2021	Prob (F-statistic):	0.00			
Time:	23:11:09	Log-Likelihood:	-3.0027e+05			
No. Observations:	21613	AIC:	6.005e+05			
Df Residuals:	21611	BIC:	6.006e+05			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
Intercept	-4.358e+04	4402.690	-9.899	0.000	-5.22e+04	-3.5e+04
sqft_living	280.6236	1.936	144.920	0.000	276.828	284.419
Omnibus:	14832.4	Durbin-Watson:	1.983			
Prob(Omnibus):	0.0	Jarque-Bera (JB):	546444.709			
Skew:	2.8	Prob(JB):	0.00			
Kurtosis:	26.9	Cond. No.	5.63e+03			

1. How do we determine the coefficients?
2. How well does the model fit?
3. How significant are the coefficients?
4. How well does the model predict on unseen data?

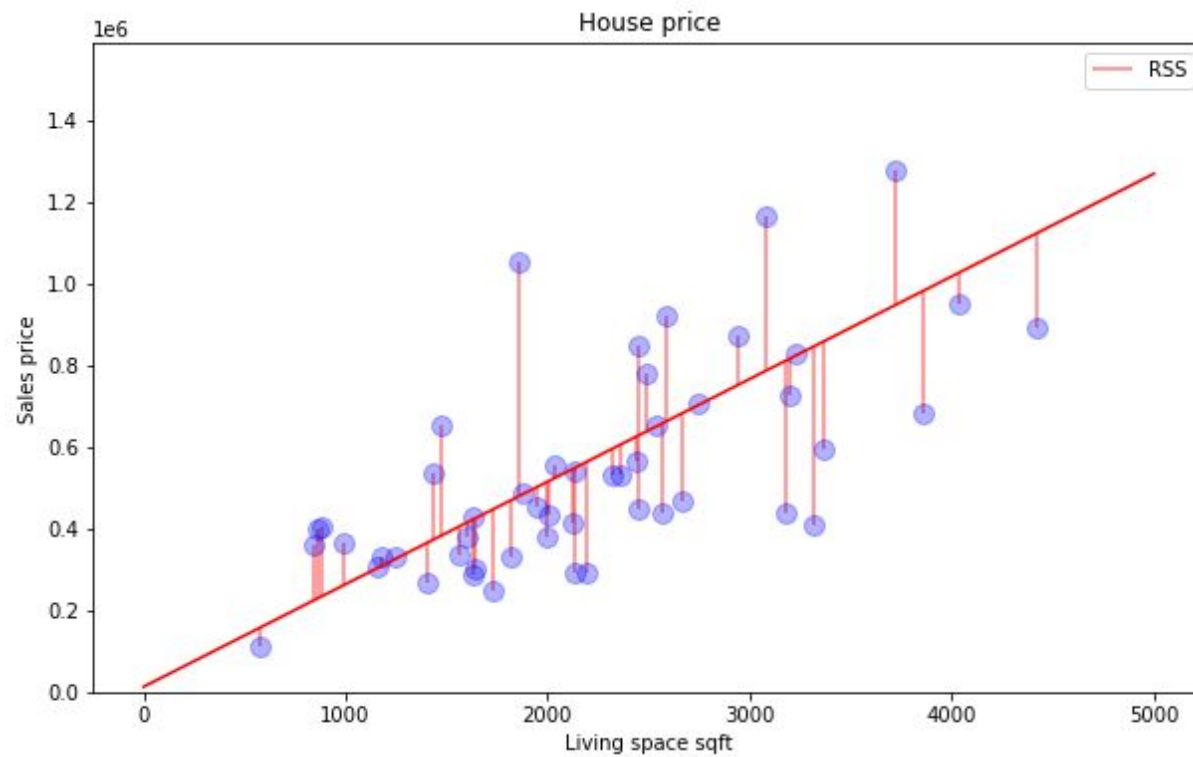
Q1. How do we find the coefficients?

$$Y_i = \underbrace{\beta_0 + \beta_1 X_i}_{\text{Coefficients}} + \underbrace{\epsilon_i}_{\text{Residual}}$$

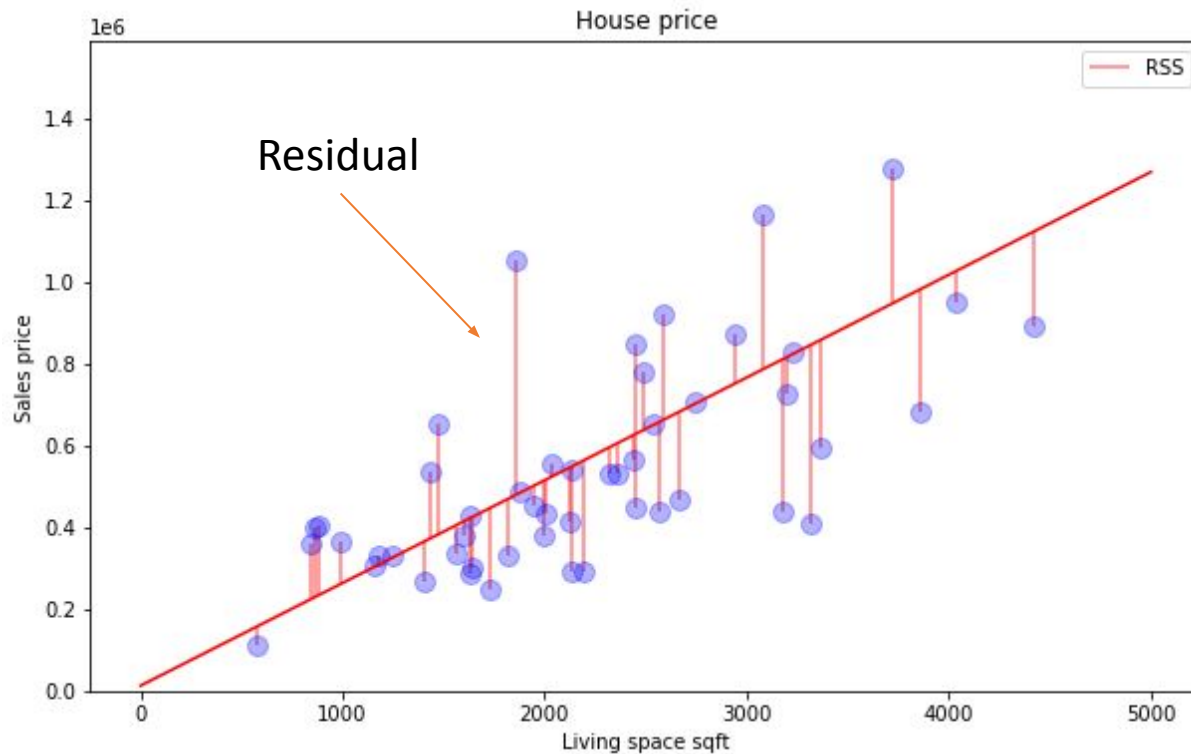
$\hat{y}_i$

The diagram shows the linear regression equation $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$. A light blue rectangular box highlights the term $\beta_0 + \beta_1 X_i$. Below this box, the word "Coefficients" is written in blue, with two blue arrows pointing upwards to β_0 and β_1 . To the right of the box, the term ϵ_i is underlined in blue, with the word "Residual" written in blue below it. A blue arrow points from the top of the box to the symbol $\hat{y}_i$ located above and to the right of the box.

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$



$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$



Mean Absolute Error (MAE)

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Mean Percent Absolute Error (MAPE)

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

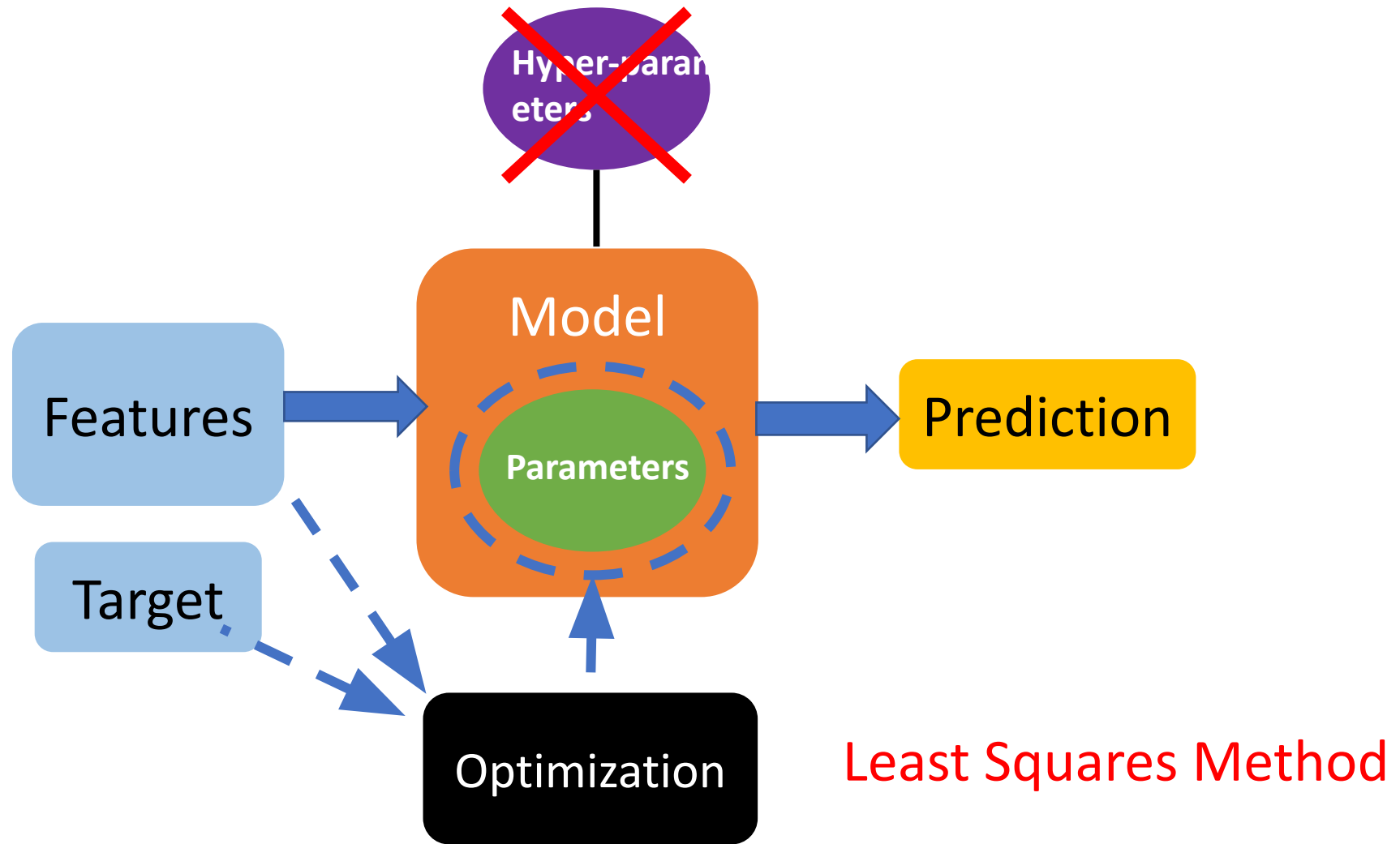
Mean Squared Error (MSE)

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Root Mean Squared Error (RMSE)

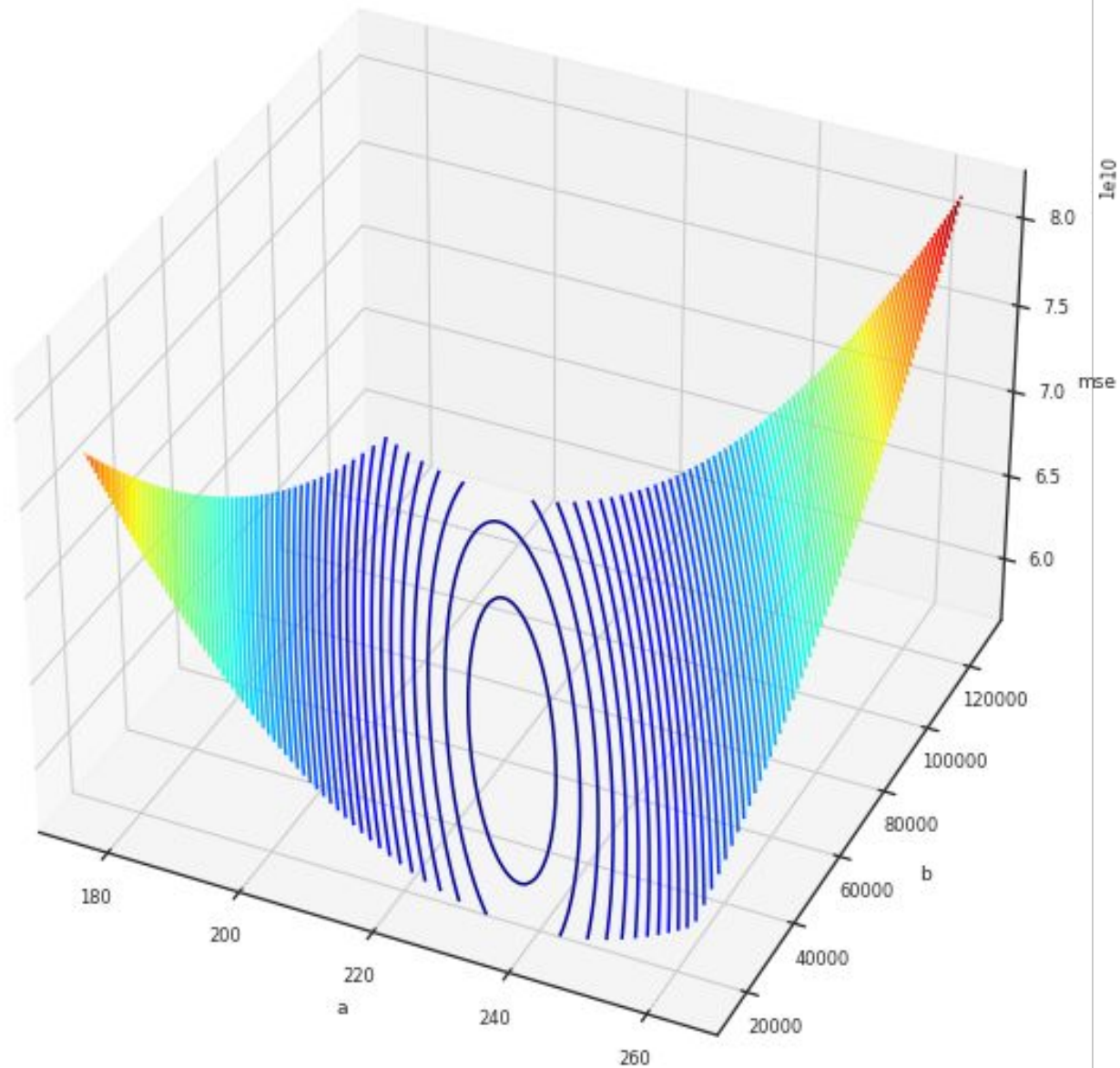
$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Optimization in Linear Regression

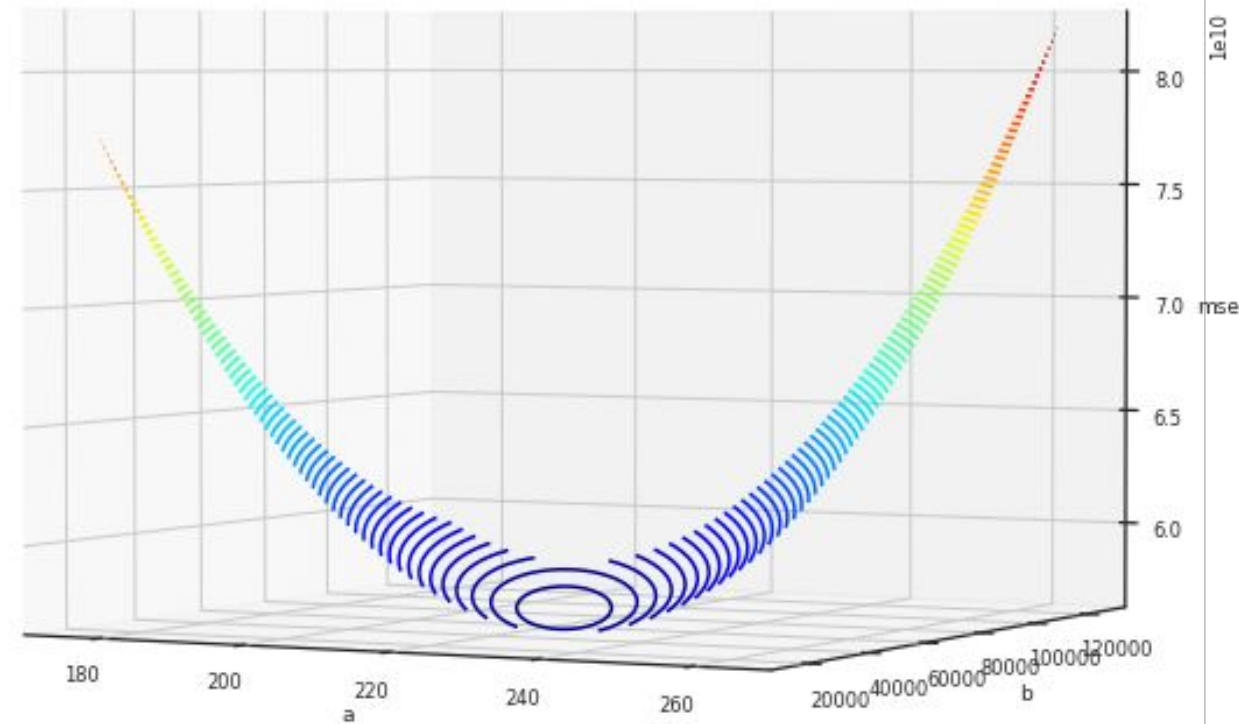


Error surface using MSE

Error surface $y=ax+b$

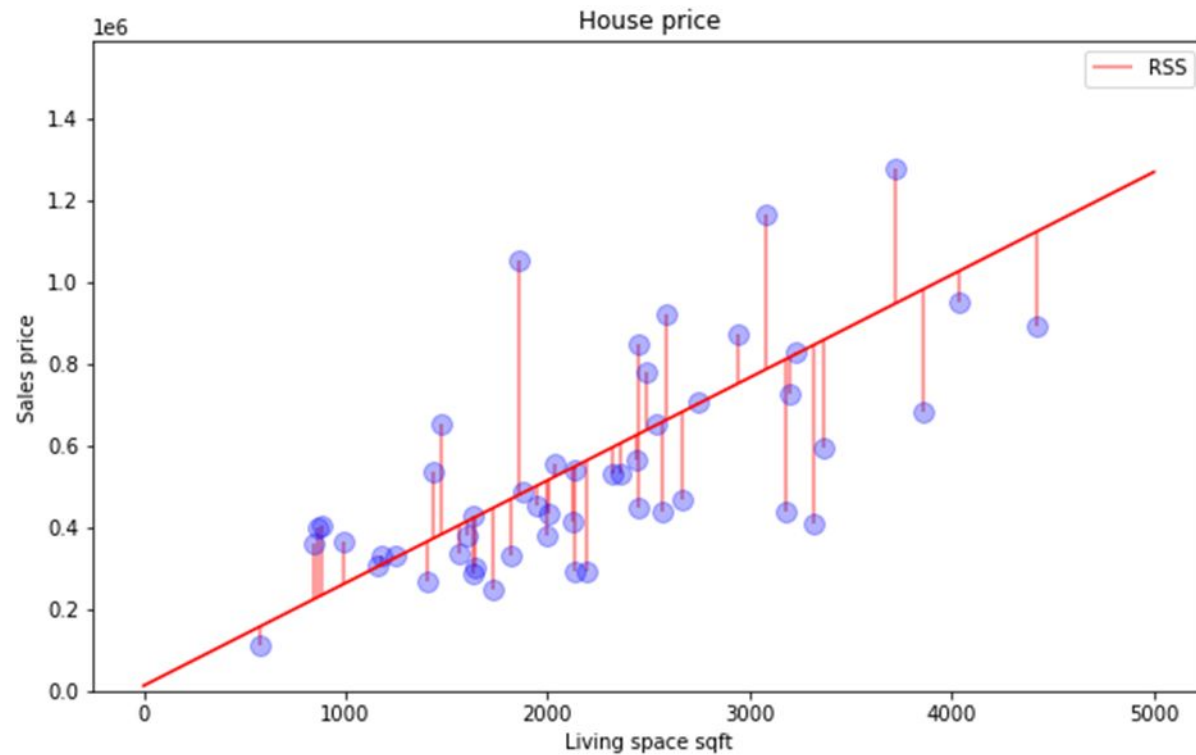


Error surface $y=ax+b$



Least Squares Method

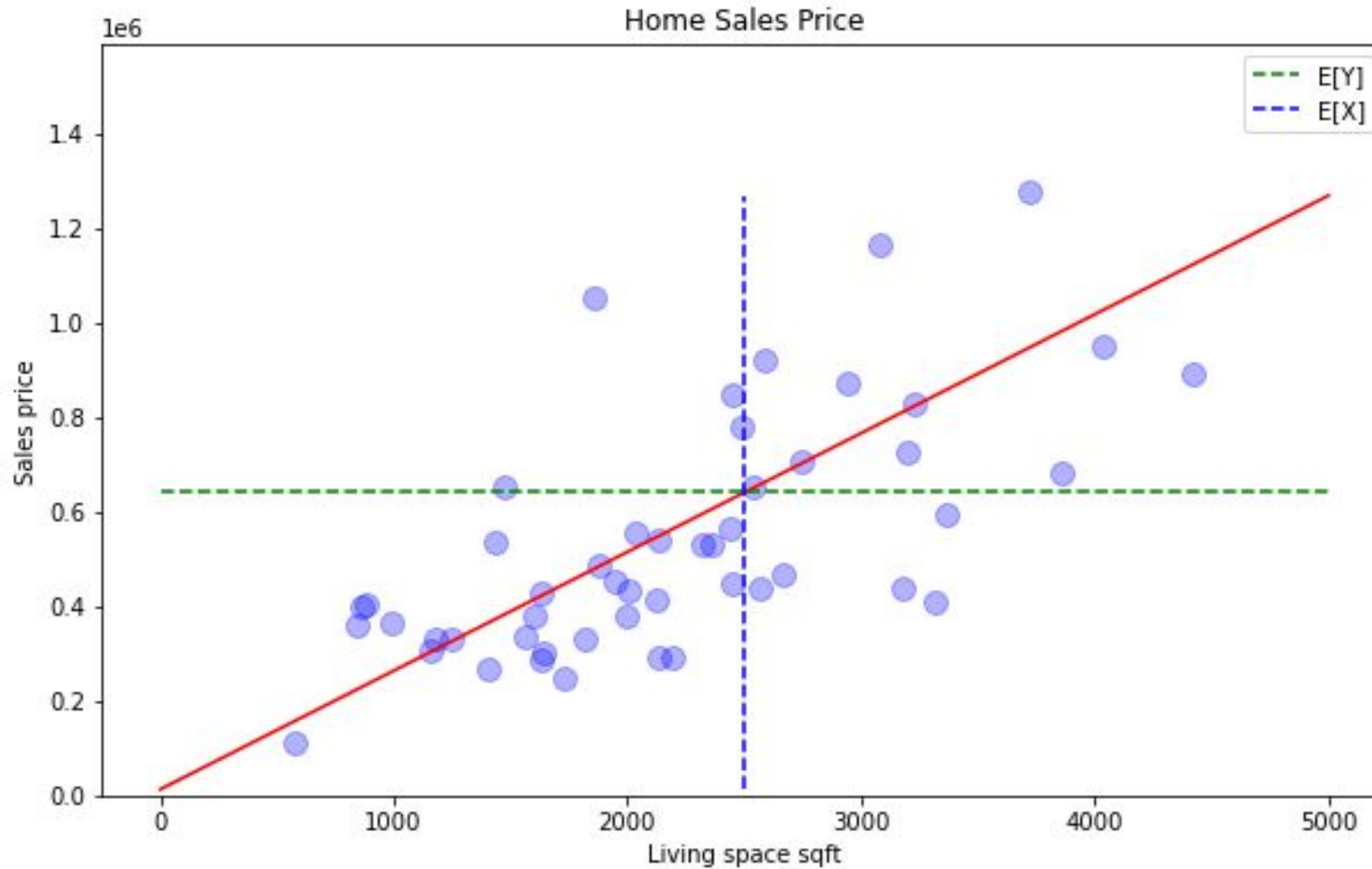
$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$



$$\beta_1 = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

$$\beta_0 = E[Y] - \frac{\text{Cov}(X, Y)}{\text{Var}(X)} E[X]$$

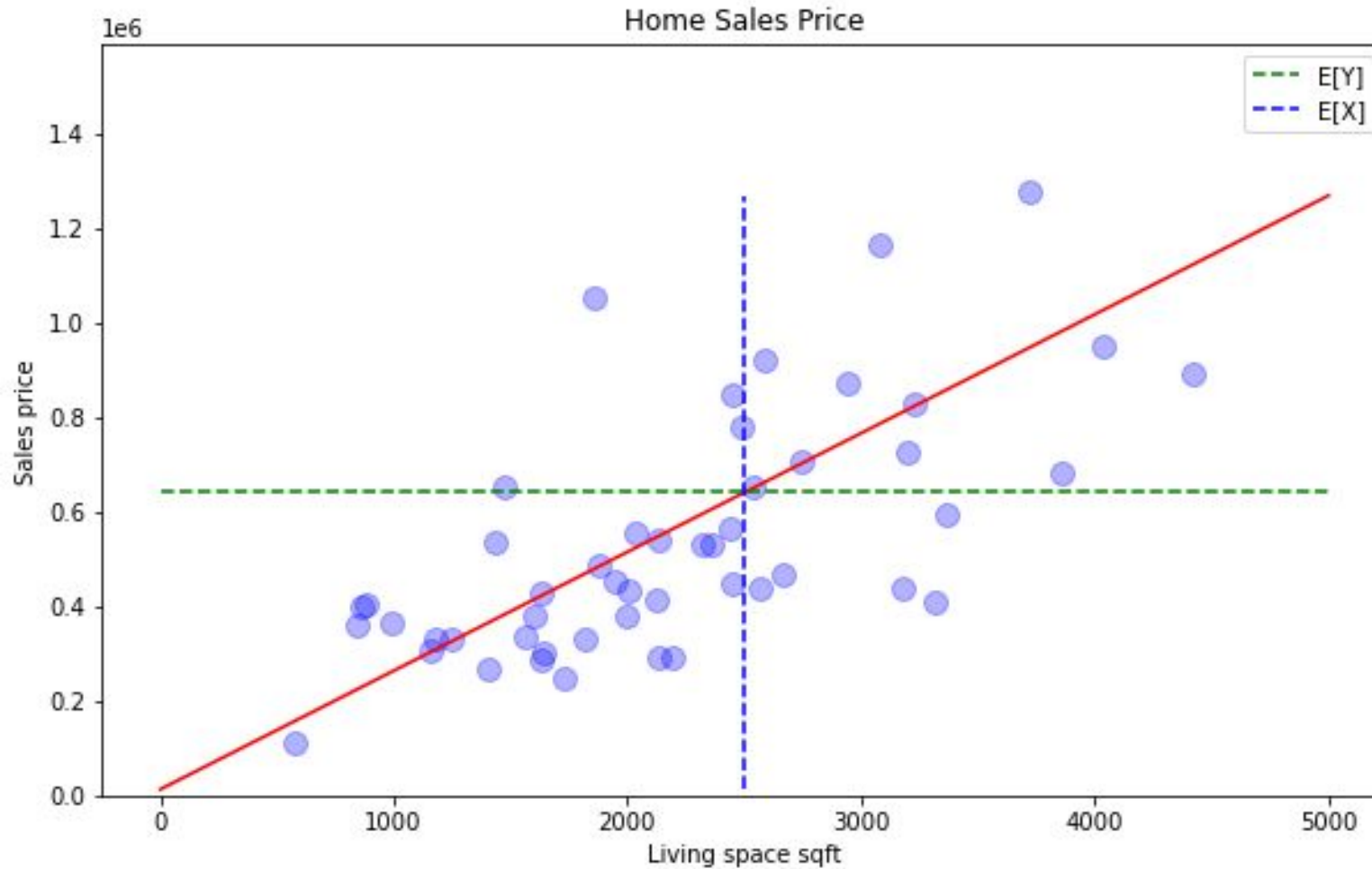
Least Squares Method



$$\beta_1 = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

$$\beta_0 = E[Y] - \frac{\text{Cov}(X, Y)}{\text{Var}(X)} E[X]$$

What happens when scaling variables?



$$\beta_1 = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

$$\beta_0 = E[Y] - \frac{\text{Cov}(X, Y)}{\text{Var}(X)}E[X]$$

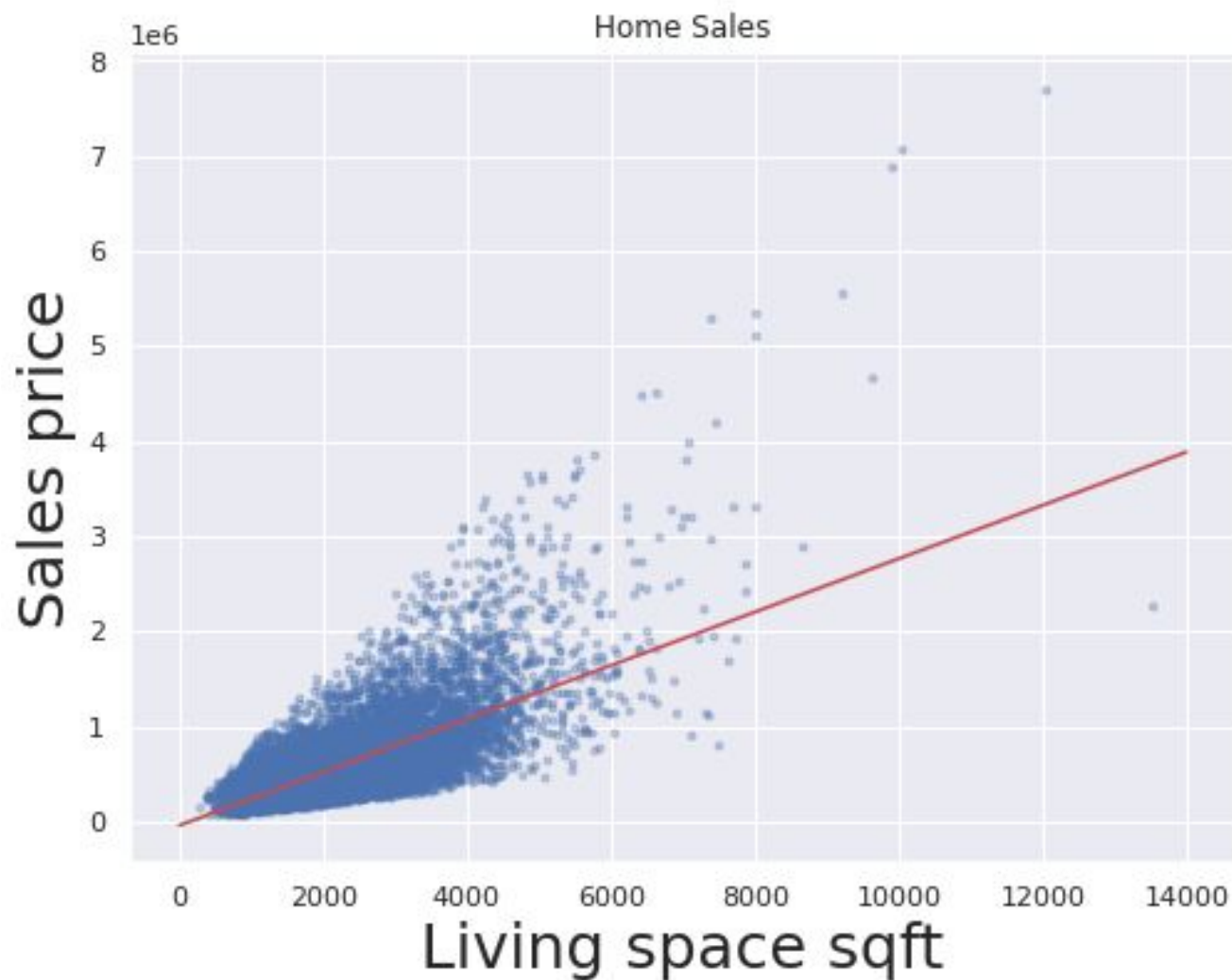
Least Squares Method in Multivariate case

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \mathbf{X} = \begin{bmatrix} 1 & X_{11} & \cdots & X_{1p} \\ 1 & X_{21} & \cdots & X_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & \cdots & X_{np} \end{bmatrix}, \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}$$

$$\frac{\partial \text{MSE}}{\partial \boldsymbol{\beta}} = 2\mathbf{X}^\top \mathbf{X} \boldsymbol{\beta} - 2\mathbf{X}^\top \mathbf{Y} + \mathbf{0} = \mathbf{0}$$

$$\boldsymbol{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$$

Q2. How well does the model fit?



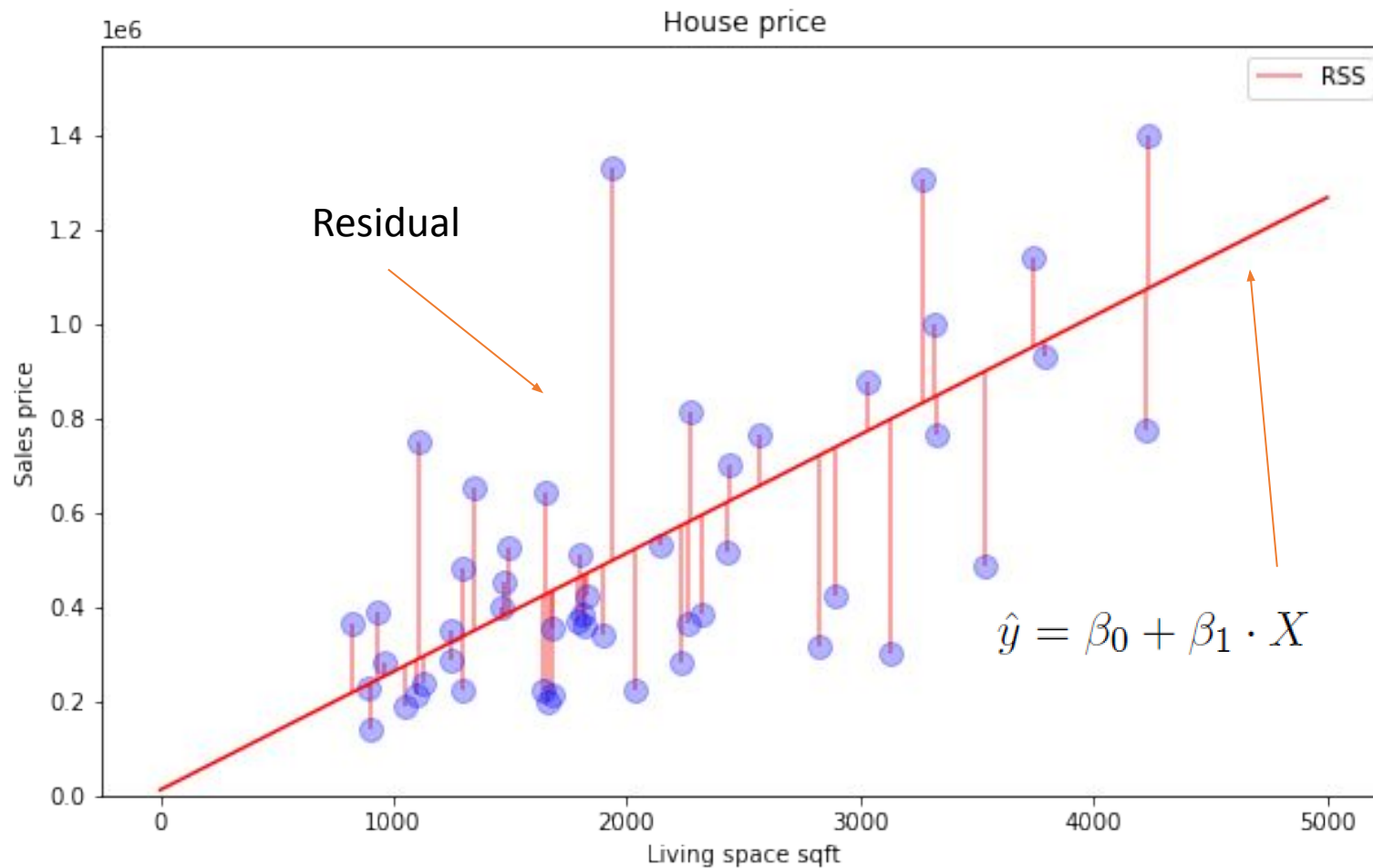
OLS Regression Results

Dep. Variable:	price	R-squared:	0.493
Model:	OLS	Adj. R-squared:	0.493
Method:	Least Squares	F-statistic:	2.100e+04
Date:	Thu, 25 Feb 2021	Prob (F-statistic):	0.00
Time:	23:11:09	Log-Likelihood:	-3.0027e+05
No. Observations:	21613	AIC:	6.005e+05
Df Residuals:	21611	BIC:	6.006e+05
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
Intercept	-4.358e+04	4402.690	-9.899	0.000	-5.22e+04	-3.5e+04
sqft_living	280.6236	1.936	144.920	0.000	276.828	284.419

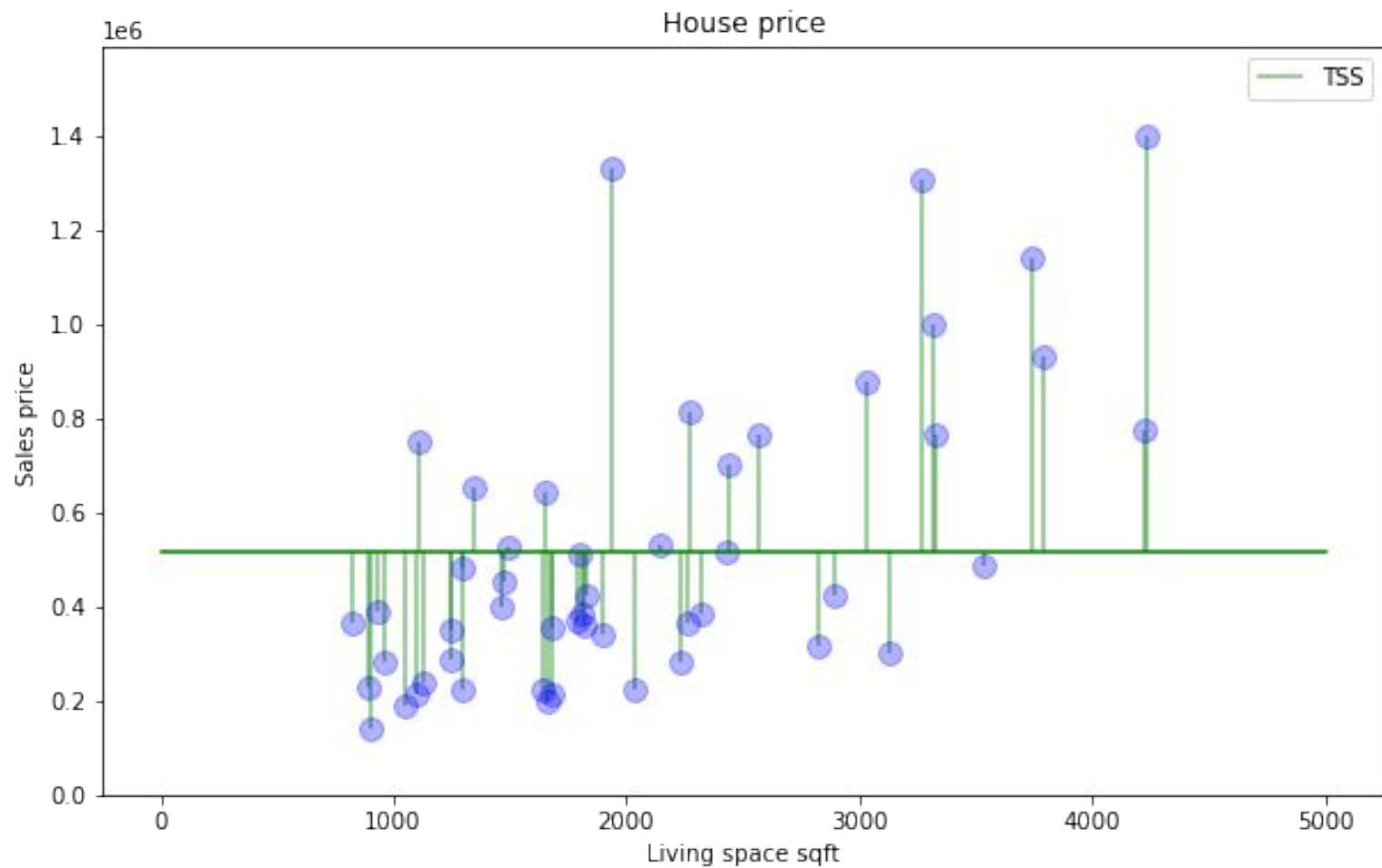
Omnibus:	14832.490	Durbin-Watson:	1.983
Prob(Omnibus):	0.000	Jarque-Bera (JB):	546444.709
Skew:	2.824	Prob(JB):	0.00
Kurtosis:	26.977	Cond. No.	5.63e+03

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$



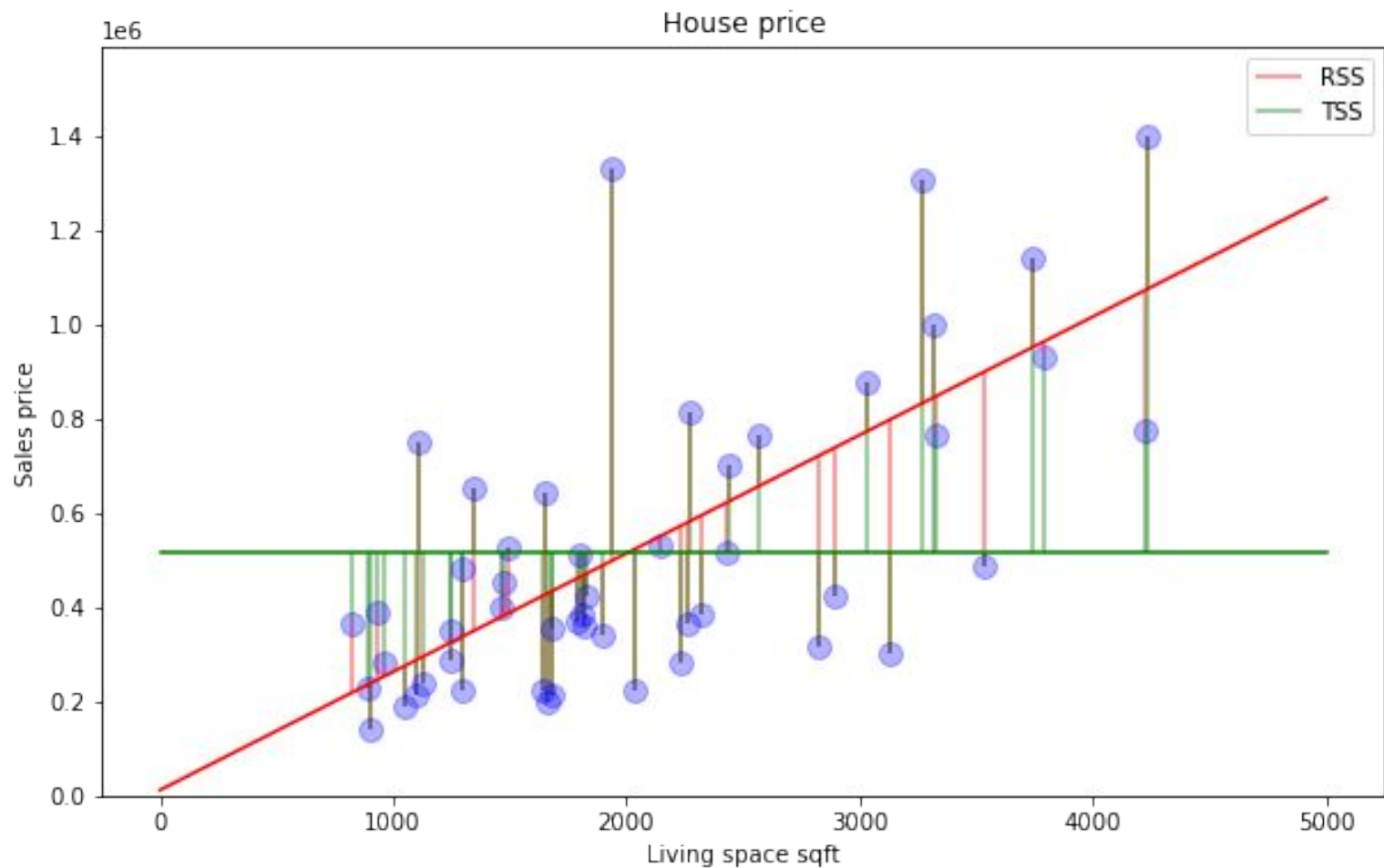
$$RSS = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

"Residual Sum of Squares"



$$\text{TSS} = \sum_{i=1}^n (y_i - \bar{y})^2$$

“Total Sum of Squares”



$$\text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$\text{TSS} = \sum_{i=1}^n (y_i - \bar{y})^2$$

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}}$$

When there is no intercept

OLS Regression Results

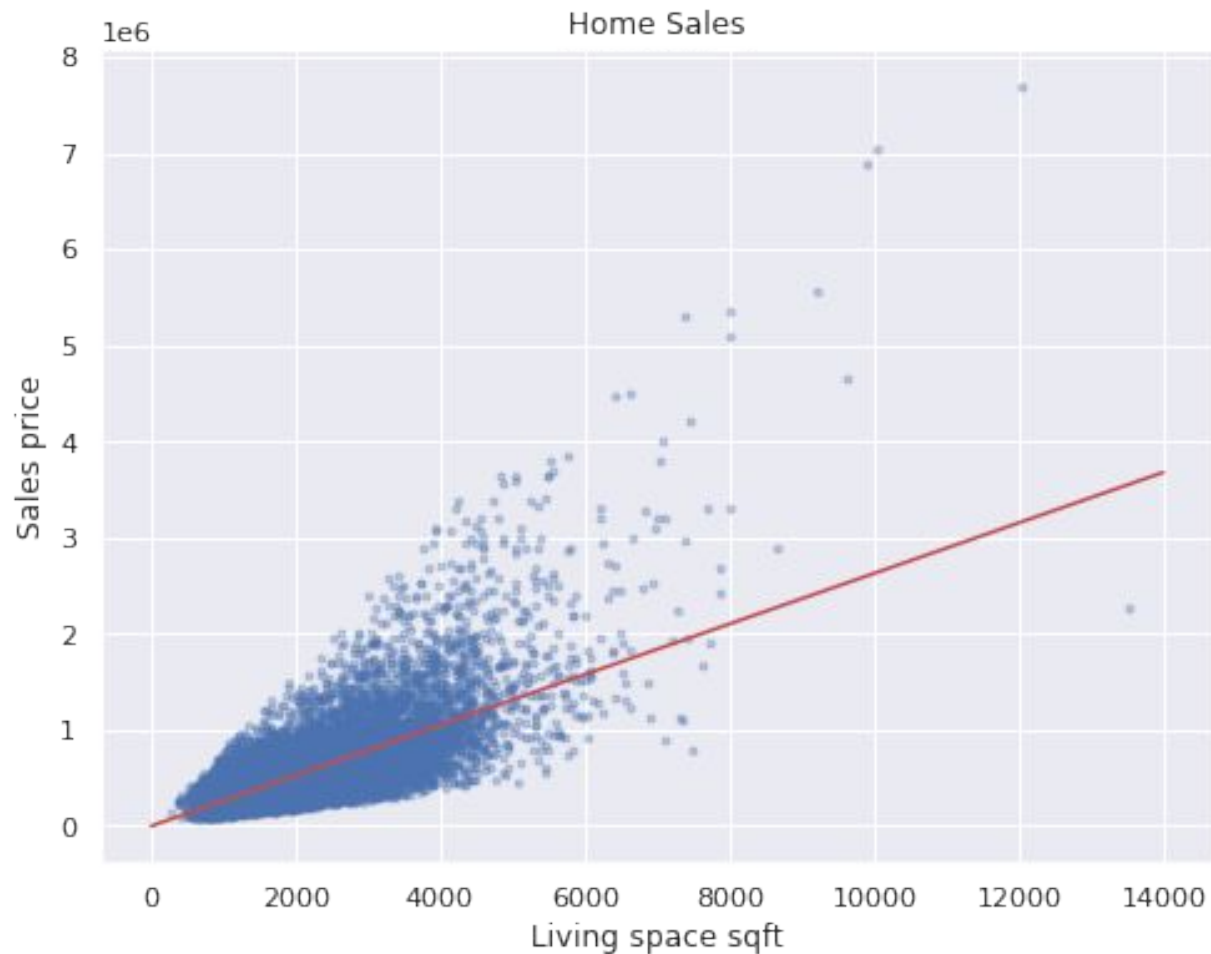
Dep. Variable:	price	R-squared:	0.493
Model:	OLS	Adj. R-squared:	0.493
Method:	Least Squares	F-statistic:	2.100e+04
Date:	Thu, 25 Feb 2021	Prob (F-statistic):	0.00
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Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
Intercept	-4.358e+04	4402.690	-9.899	0.000	-5.22e+04	-3.5e+04
sqft_living	280.6236	1.936	144.920	0.000	276.828	284.419

Omnibus:	14832.490	Durbin-Watson:	1.983
Prob(Omnibus):	0.000	Jarque-Bera (JB):	546444.709
Skew:	2.824	Prob(JB):	0.00
Kurtosis:	26.977	Cond. No.	5.63e+03



When there is no intercept



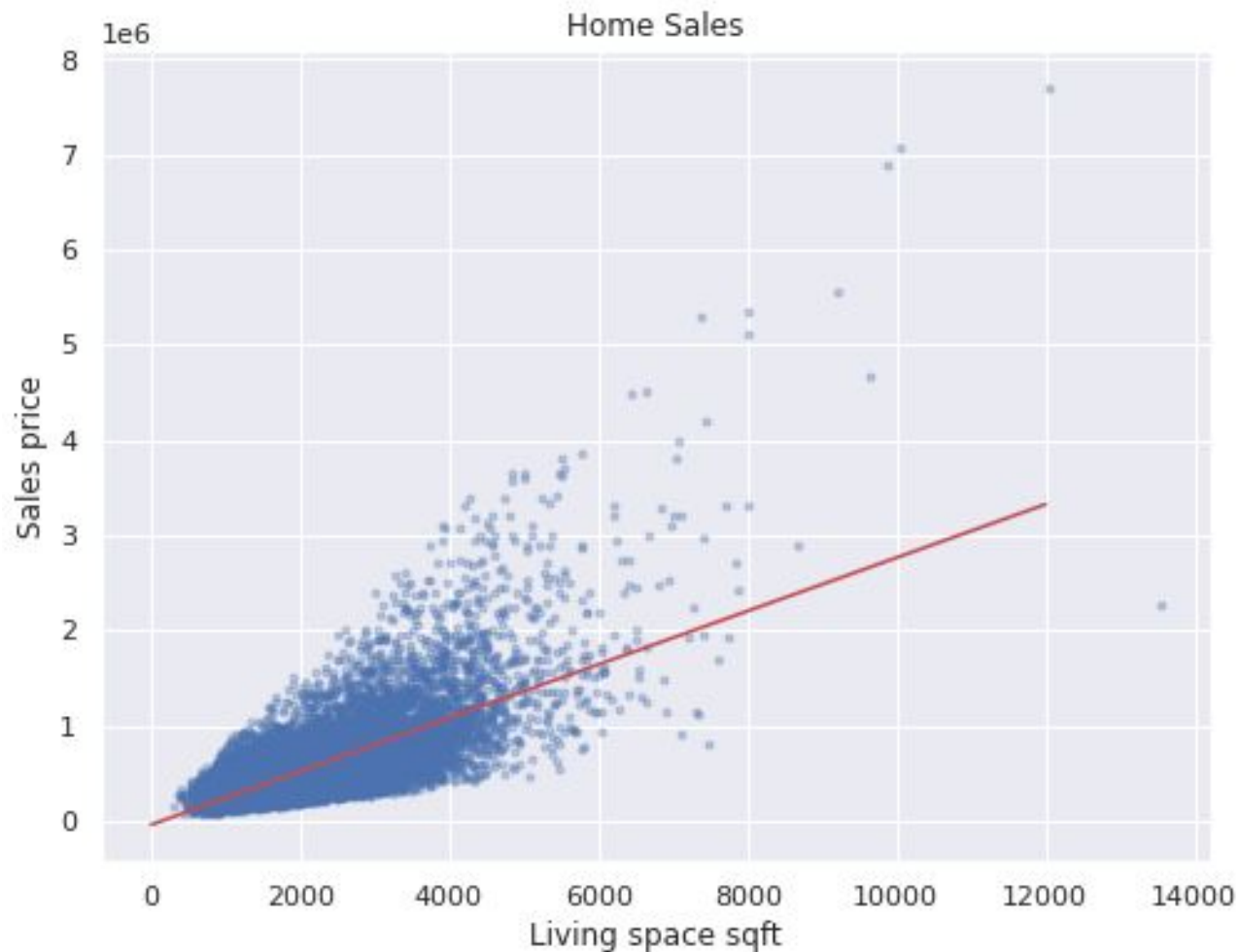
OLS Regression Results

Dep. Variable:	price	R-squared (uncentered):	0.839
Model:	OLS	Adj. R-squared (uncentered):	0.839
Method:	Least Squares	F-statistic:	1.126e+05
Date:	Thu, 25 Feb 2021	Prob (F-statistic):	0.00
Time:	23:09:13	Log-Likelihood:	-3.0032e+05
No. Observations:	21613	AIC:	6.006e+05
Df Residuals:	21612	BIC:	6.006e+05
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
sqft_living	263.0892	0.784	335.597	0.000	261.553	264.626

Omnibus:	16043.334	Durbin-Watson:	1.980
Prob(Omnibus):	0.000	Jarque-Bera (JB):	692411.844
Skew:	3.130	Prob(JB):	0.00
Kurtosis:	30.013	Cond. No.	1.00

Q3. How significant are the estimated coefficients?



OLS Regression Results

Dep. Variable:	price	R-squared:	0.493
Model:	OLS	Adj. R-squared:	0.493
Method:	Least Squares	F-statistic:	2.100e+04
Date:	Thu, 25 Feb 2021	Prob (F-statistic):	0.00
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Omnibus:	14832.490	Durbin-Watson:	1.983
Prob(Omnibus):	0.000	Jarque-Bera (JB):	546444.709
Skew:	2.824	Prob(JB):	0.00
Kurtosis:	26.977	Cond. No.	5.63e+03

From homoscedasticity assumption

$$\varepsilon \sim N(0, \sigma^2)$$

$$\text{Var}(\hat{\beta}) = (\mathbf{X}^T \mathbf{X})^{-1} \sigma^2$$

$$\hat{\sigma}^2 = \frac{1}{N - p - 1} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

$$\text{SE}(\hat{\beta}_0)^2 = \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right]$$

$$\text{SE}(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Bootstrap

p-value

Null hypothesis

$$H_0 : \beta_1 = 0$$

t-statistics

$$t = \frac{\hat{\beta}_1 - 0}{\text{SE}(\hat{\beta}_1)}$$

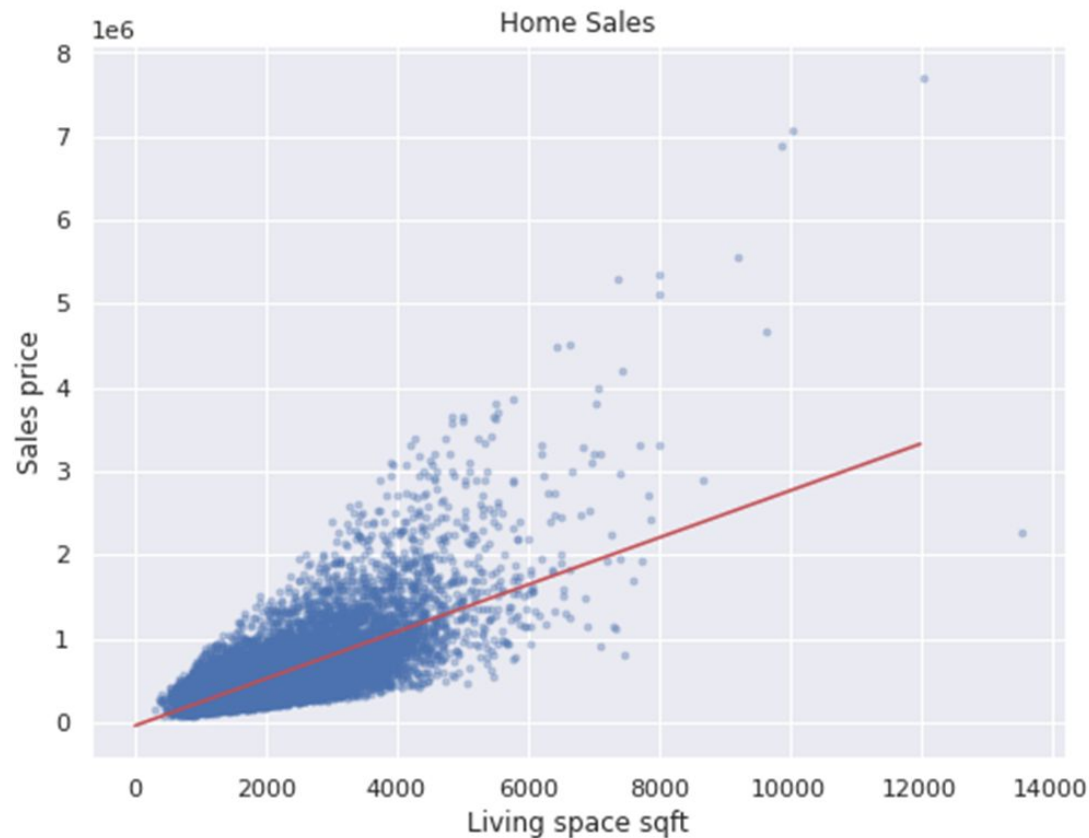
Alternative hypothesis

$$H_A : \beta_1 \neq 0$$

p-value

	coef	std err	t	P> t	[0.025	0.975]
Intercept	-4.358e+04	4402.690	-9.899	0.000	-5.22e+04	-3.5e+04
sqft_living	280.6236	1.936	144.920	0.000	276.828	284.419

Confidence Interval for Coefficients



OLS Regression Results

Dep. Variable:	price	R-squared:	0.493
Model:	OLS	Adj. R-squared:	0.493
Method:	Least Squares	F-statistic:	2.100e+04
Date:	Thu, 25 Feb 2021	Prob (F-statistic):	0.00
Time:	23:11:09	Log-Likelihood:	-3.0027e+05
No. Observations:	21613	AIC:	6.005e+05
Df Residuals:	21611	BIC:	6.006e+05

Df Model:	1
Covariance Type:	nonrobust

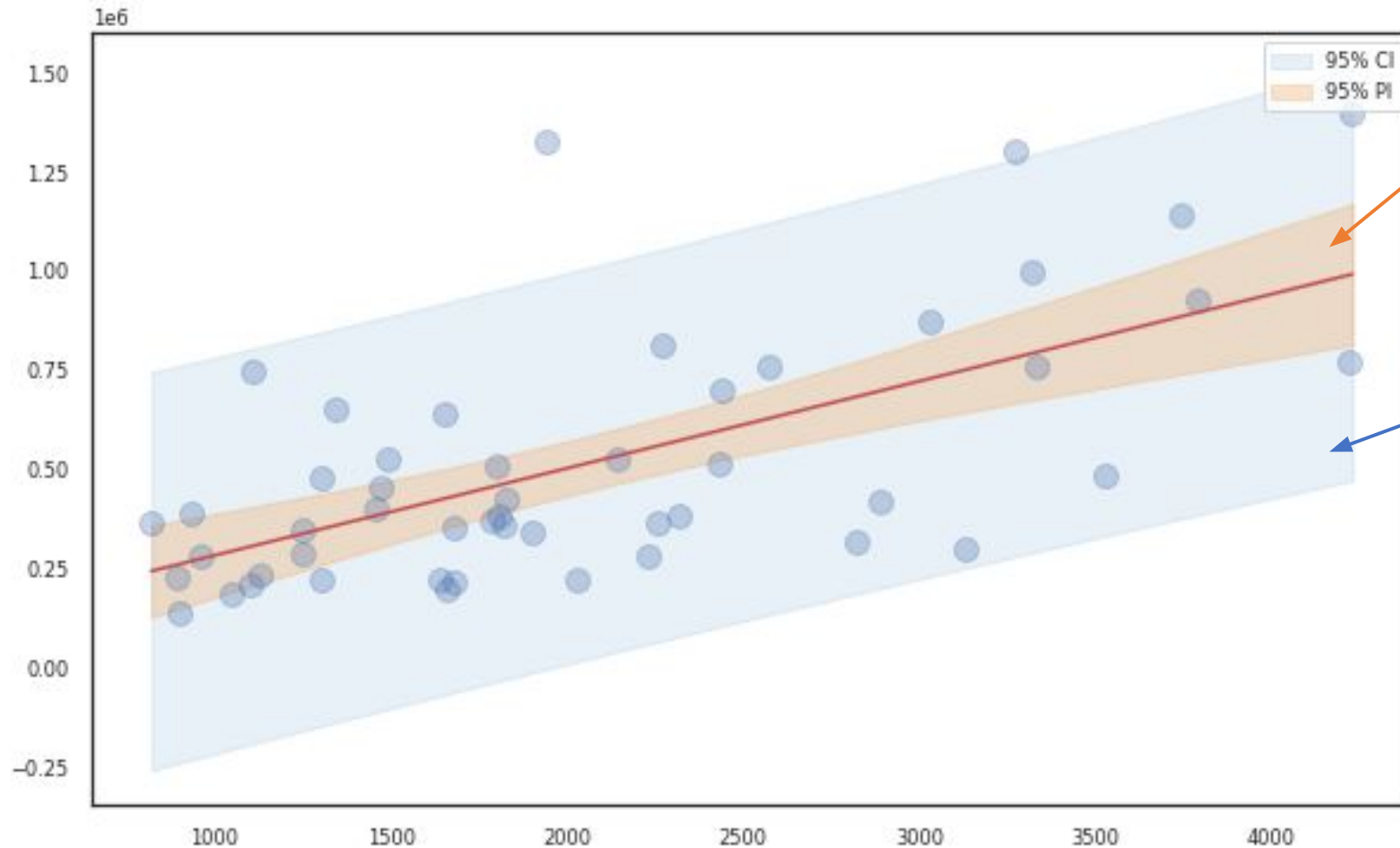
95% CI for the coefficients

	coef	std err	t	P> t	[0.025	0.975]
Intercept	-4.358e+04	4402.690	-9.899	0.000	-5.22e+04	-3.5e+04
sqft_living	280.6236	1.936	144.920	0.000	276.828	284.419

Omnibus: 1 $\left[\hat{\beta}_1 - 2 \cdot \text{SE}(\hat{\beta}_1), \hat{\beta}_1 + 2 \cdot \text{SE}(\hat{\beta}_1) \right]$
Prob(Omnibus):

Skew:	2.824	Prob(JB):	0.00
Kurtosis:	26.977	Cond. No.	5.63e+03

Confidence Intervals for Regression



**95% Confidence Interval
(for the regression line)**

**95% Prediction Interval
(for sample points)**

Q4.How well does the model predict on unseen data?

Popular Error metrics

Mean Absolute Error (MAE)

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Mean Percent Absolute Error (MAPE)

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

Mean Squared Error (MSE)

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Root Mean Squared Error (RMSE)

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Dataset	R2	MSE	MAPE
Train (80%)	0.492	6.632 E10	0.3598
Test (20%)	0.494	7.648 E10	0.3570

Summary

1. How do we determine the coefficients?
2. How well does the model fit?
3. How significant are the coefficients?
4. How well does the model predict on unseen data?